

SECTION – A

Q1 (Compulsory) – 12 Marks

a) ANY THREE ($3 \times 2 = 6$)

1. What is Algorithm?
2. Define Dynamic Memory Allocation.
3. What is Stack Overflow?
4. Define Dequeue.

b) ANY THREE ($3 \times 2 = 6$)

1. Define Graph.
 2. What is Hash Function?
 3. Define Recursion.
 4. What is Load Factor?
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Q2 – 9 Marks

a) Explain Queue operations. (4)

b) ANY ONE (5)

- Explain Circular Queue with diagram.
 - Explain Priority Queue.
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Q3 – 9 Marks

a) Explain Merge Sort. (4)

b) ANY ONE (5)

- Sort using Merge Sort: 67, 25, 89, 12
- Explain time complexity of Merge Sort.

SECTION – B

Q4 (Compulsory) – 12 Marks

a) ANY THREE ($3 \times 2 = 6$)

1. Define Tree.
2. What is Balance Factor?
3. Define Hash Table.
4. What is DFS?

b) ANY THREE ($3 \times 2 = 6$)

1. Define BFS.
 2. What is BST?
 3. Define Adjacency Matrix.
 4. What is Collision Resolution?
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Q5 – 9 Marks

- a) Explain AVL Tree rotations. (4)
- b) ANY ONE (5)

- Insert 10, 20, 30 into AVL Tree.
 - Explain LL and RR rotations.
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Q6 – 9 Marks

- a) Explain BFS traversal. (4)
- b) ANY ONE (5)

- Explain DFS traversal.
- Compare BFS and DFS.